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Simplex Work Test

April 7, 2026

Abstract

Provide a concise summary of the research question, method, key results, and primary conclusion.

1 Introduction

Transformers have been shown to contain linearaly transformed representations of the belief state geometry in their residual streams. Real data-generating processes can often be at least approximately factored. For example, consider some data obtained from human-written text. One factor might be the communicative intentions of the author. Another factor might be whether or not they make a typo on the current word. These two hidden states jointly inform the content of each given word that will be generated, but approximately do not affect each other. It has been shown in much simpler cases that that the belief state geometries of the individual factors are repsented in the residual stream in orthogonal subspaces.

In this work test, we are studying a different situation where a data-generating process factors into different process which are mutually exclusively used to generate different datapoints. For example, a dataset which contains text from the internet will have different datapoints generated by different authors, who can be modelled in the data generating process as independent factors.

In particular we describe the data generating process having $d = \sum_{n=1}^{N} d_n$ hidden states and a vocabulary of tokens $\mathcal{X}$ which each of N factors may emit from. The underlying process for generating data is completely described by the generalised hidden markov model with latent space

$$\mathbb{R}^d = \bigoplus_{n=1}^{N} \mathbb{R}^{d_n}.$$

Given an inital state vector $\eta^{\varnothing}$ in the latent space, operators $T^{(x)}$ encoding the ground truth transition probability of the hidden state upon seeing the token $x \in \mathcal{X}$, and the vector $\mathbf{1}$ chosen to integrate these probabilites, the probability of the the sequence of tokens $x_1, x_2, \ldots, x_\ell$ is equal to

$$\eta^{\varnothing} T^{(x_1)} T^{(x_2)} \ldots T^{(x_\ell)} \mathbf{1}.$$

This generalised hidden markov model describes the belief state geometry, that is, the process for how the probabiltiy of being in a given hidden state evolves upon observing additional tokens.

Predictions

Note that this analysis was done before any experiments, as is evident in github:

Mathematical Considerations

We have two many hypotheses for what the geometry could look like, which I will now describe.

In this specific case of this research, the transition operators $T^{(x)}$ are block diagonal operators assembled from transition operators $T_n^{(x)}$ corresponding to each factor n.

Since each sequence is mandated to be generated by a single factor, the probability simplex Δ^{d-1} which is the space that this generalised hidden markov process occurs on can be factored as the tensor product $\Delta^{N-1} \otimes \left(\bigotimes_{n=1}^{N} \Delta^{d_n-1}\right)$ where the first coefficient corresponds to the probability distribution over the independent factors and the later terms correspond to the probability distribution over hidden states within each factor.

Each of these N generalised hidden Markov models describe a conditional belief state geometry; they give the conditional probability of the process being in a hidden state given it is in a particular factor. Importantly, the probabiity distribution over the factors is should a GHMM in the same way because computing the transition probabilites of this distribution at a given time requires the current probability distribution over all of the hidden states.

Candidate Geometries

Nonetheless, this gives two clear alternatives for the representation of the belief state geometry in the residual stream:

1. A linear transform of the d-dimensional simplex describing the whole data-generating progress, that is, directions represent the true probabilites of being in a given hidden state.

2. $(N+1)$ orthogonal subspaces, N of which cotnain linear transforms of the belief state geometry of the factors and the last encodes the probability distribution over which factor one is in.

Note these are distinct geometries to the point that they are not linear transforms of eachother. Thus we expect that we can empirically determine between them using linear regression.

Architectural Obstruction to Updating Belief States

A major conceptual difficulty we run into is that the residual stream at a particular layer and token can only be downstream of previous tokens that are in previous layers. That means that if the residual stream in a particular layer is the first to compute the belief state, it cannot use the computed belief state from previous tokens. Thus it is important to realise that the transformer can not just compute the update on observing the last token, but must compute the update upon having observed the entire sequence.

Because this operation is highly non-linear (since it involves multiplying a bunch of probabilies), I suggest that the model has to do these computation using log probabilites to avoid needing to do multiplications. If I am right (and not just having some fundamental conceptual

misunderstanding about transformers) I should be able to validate this experimentally by doing linear regression with the log geometries instead.

We present some intuition pumps for thinking about thse altenratives:

- based on having seen the images in previous work of the fractal-like orbits of the belief state geometry in the residual stream evolving over the course of training, and the fact that in early training steps none of the more intricate structures seems to be captured, representation (2) seems less likely because it might require the model to decouple the factors early in the process when there is too much noise.
- Given the representation (1), we are in some sense one step away from next token prediction, because we only need to do one matrix multipication between the probabilites of the hidden states and the probabilites of emitting tokens given those states. Meanwhile the representation (2) is in a sense two steps, because we first need to obtain the actual probabilites of the hidden states before we can use them to compute the next token probabilites.
- (2) the structure seems a lot more natural, in the sense that it is how a world model would most likely feel for a human and I think how most people would imagine LLM world models to work. Since the inductive biases of transformers seem to vibe with what people consider natural in some non trivial way this could be taken as some evidence.

My intuition is that a model with a single hidden layer, if it can succeed in learning the data will do so through reperesntation (1). If there are multiple layers, I have a feeling that an early layer would learn representation (2) and a later layer will use reperetnation (1) that has been computed from the representation in the early layer. With 3 or more layers I would suspect that the earliest layer might only compute the representations for the individual facotrs, then a later layer would use this to compute the probabilites of each factor, (that is, have obtained representation (2)), and an even later layer would have used this to comptue representation (1).

2 Experiments

Setup

We randomly generate three different Mess3 processes, outputting into the same vocabulary of three tokens. Then we generate a a dataset made in equal parts from sequences generated by each process, mimicking a data-generating process which randomly chooses between them.

Then we pretrain transformers on this data, primarily a model with 2 attention layers and..

Representations of the Belief State Geometry

In this section I have tried to give a natural logical progression of the investigations I did. If I should instead have given the most sohpisticated version of my method and later listed ablations, please let me know.

We follow the methodology of TODO, and use a fixed subset of the test data to generate a ground truth belief state geometry dataset for each Mess3 factor on 2-simplices, for the

probability distribution on being in each factor on 2-simplices, and the total data-generating process on an 8-simplex. Similarly, we use the train model for inference on the same sequences and capture the residual stream before the unembeddings.

We first use linear regression to find a linear map between from the latter residual stream into each of the former datasets, and record the mean squared error. As a control, we repeat this analysis with a newly intialised version of the model that shouldn't have any learned representation of the belief state geometry.

TODO

The data decisively shows that, contrary to my predictions, the belief state geometries of the factors are represented and the belief state geometry of the entire process is not. Although the belief state geometry over the factors does not show an especially big change in mean square error compared the control, figure TODO demonstrates that it is signifcantly represented.

Using log Belief State Probabilites

As mentioned in the introduction, there are mechanistic reasons for believing that the residual stream is more likely to represent the beliefs states using their log probabilites. To verify this, we do linear regression instead onto the log probabilites of the ground truth belief states, and then convert them back to probabilites before computing mean square error as a fair comparison.

This approach constitently demonstrates superority compared with directly computing the probabilites.

Orthgonality in the Residual Stream

In order to describe the subspaces of the residual stream responsible for representing the different factors, we replace the regression analysis by a reduced regression analysis which restricts the

Procedure

Detail the steps, tools, model setup, or experimental protocol.

Evaluation

Describe metrics, baselines, and statistical tests if applicable.

3 Results

Present core findings clearly.

Primary Findings

Summarize the main quantitative or qualitative outcomes.

Additional Analyses

Include robustness checks, ablations, subgroup analysis, or error analysis.

4 Discussion

Interpret the results, limitations, and implications.

Interpretation

Explain what the findings mean in relation to the research question.

Limitations

Note constraints that affect validity or generalizability.

Future Work

List practical next steps and open research directions.

Acknowledgments

Optional.

References